Basics Physics-Note Curriculum

One hundred planned prerequisite notes

Thirty standalone textbook drafts plus seventy reserved curriculum scaffolds

Curriculum status. Thirty numbered notes are substantive standalone textbook drafts, version 1.0. Batch 1 comprises Basics 01, 02, 03, 05, 06, 07, 11, 12, 13, and 17. Batch 2 comprises Basics 18, 19, 21, 22, 24, 25, 26, 29, 31, and 32. Batch 3 comprises Basics 34, 35, 36, 37, 39, 40, 41, 42, 44, and 48. The remaining seventy notes are explicitly labeled curriculum scaffolds. Draft status means that derivations, interpretation, a computational laboratory, exercises, symbols, and references are present; it does not imply peer review or final textbook completeness.

Contents

1 How to use this curriculum

The notes form a prerequisite ladder. A student may follow the complete order from Basics 01 through Basics 100 or use the prerequisite tables in advanced module notes. Every basics note remains a separate document; no combined course-volume PDF is maintained. The status printed beside every entry is authoritative for this repository release.

2 Completed standalone batches

Batch 1: mathematical, electromagnetic, and introductory-plasma foundations:

$$\{1, 2, 3, 5, 6, 7, 11, 12, 13, 17\}.$$

Batch 2: plasma collectivity, kinetic moments, ideal MHD, linearization, and elementary Alfvén-wave physics:

$$\{18, 19, 21, 22, 24, 25, 26, 29, 31, 32\}.$$

Batch 3: field-line propagation, Alfvén continua and gap modes, wave-particle resonance, kinetic phase space, guiding-center motion, and energetic-particle distributions:

$$\{34, 35, 36, 37, 39, 40, 41, 42, 44, 48\}.$$

Together, these thirty notes provide a path from mathematical fields and Maxwell equations through ideal-MHD Alfvén waves and into the reduced kinetic physics required by Modules 3–5.

3 I - Mathematical and computational foundations

01. Basics 01: Units, Dimensions, and Normalization
Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

02. Basics 02: Vectors, Fields, and Coordinate Systems
Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

03. Basics 03: Vector Calculus for Plasma Physics
Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

04. Basics 04: Ordinary Differential Equations
Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

05. Basics 05: Partial Differential Equations
Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

06. Basics 06: Linear Algebra and Eigenvalue Problems
Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

07. Basics 07: Fourier Analysis and Normal Modes
Textbook draft v1. *Primary connections: All modules, especially Modules 4, 8, 9, and 11.*

08. Basics 08: Complex Numbers, Phasors, and Wave Notation
Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

09. Basics 09: Variational Principles and Energy Methods
Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

10. Basics 10: Probability, Statistics, and Monte Carlo
Curriculum scaffold. Primary connections: All modules, especially Modules 4, 8, 9, and 11.

4 II - Electromagnetism foundations

11. Basics 11: Electric and Magnetic Fields
Textbook draft v1. *Primary connections: Modules 1, 3, 4, 5, and 8.*

12. Basics 12: Maxwell Equations
Textbook draft v1. *Primary connections: Modules 1, 3, 4, 5, and 8.*

13. Basics 13: Lorentz Force and Charged-Particle Motion
Textbook draft v1. *Primary connections: Modules 1, 3, 4, 5, and 8.*

14. Basics 14: Electromagnetic Energy and Poynting Theorem
Curriculum scaffold. Primary connections: Modules 1, 3, 4, 5, and 8.

15. Basics 15: Magnetic Pressure, Tension, and Flux
Curriculum scaffold. Primary connections: Modules 1, 3, 4, 5, and 8.

16. Basics 16: Gauge Fields and Electromagnetic Potentials
Curriculum scaffold. Primary connections: Modules 1, 3, 4, 5, and 8.

5 III - Elementary plasma physics

6 IV - Magnetohydrodynamics

7 V - Plasma waves and Alfvén physics

31. Basics 31: Introduction to Waves in Plasmas
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

32. Basics 32: Alfvén Waves from First Principles
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

33. Basics 33: Fast and Slow Magnetosonic Waves
Curriculum scaffold. Primary connections: Modules 4, 5, and 6.

34. Basics 34: Parallel Wave Number and Field-Line Propagation
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

35. Basics 35: Alfvén Continuum
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

36. Basics 36: Coupled Harmonics and Avoided Crossings
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

37. Basics 37: Toroidal Alfvén Eigenmodes
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

38. Basics 38: Stellarator Alfvén Eigenmodes
Curriculum scaffold. Primary connections: Modules 4, 5, and 6.

39. Basics 39: Wave-Particle Resonance
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

40. Basics 40: Growth, Damping, and Complex Frequency
Textbook draft v1. *Primary connections: Modules 4, 5, and 6.*

8 VI - Kinetic theory and energetic particles

41. Basics 41: Phase Space and Distribution Functions
Textbook draft v1. *Primary connections: Modules 2, 3, 5, and 6.*

42. Basics 42: Vlasov Equation
Textbook draft v1. *Primary connections: Modules 2, 3, 5, and 6.*

43. Basics 43: Boltzmann and Fokker-Planck Equations
Curriculum scaffold. Primary connections: Modules 2, 3, 5, and 6.

44. Basics 44: Guiding-Center Theory
Textbook draft v1. *Primary connections: Modules 2, 3, 5, and 6.*

9 VII - Fusion and burning-plasma foundations

10 VIII - Stellarator and magnetic-confinement foundations

59. Basics 59: Magnetic-Confinement Concepts
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

60. Basics 60: Tokamaks and Stellarators
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

61. Basics 61: Toroidal Coordinates and Magnetic Surfaces
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

62. Basics 62: Rotational Transform, Magnetic Shear, and Resonances
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

63. Basics 63: Stellarator Symmetry and Fourier Geometry
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

64. Basics 64: Quasisymmetry and Neoclassical Confinement
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

65. Basics 65: Magnetic-Field Coils and the Biot-Savart Law
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

66. Basics 66: MHD Equilibrium Codes and VMEC Concepts
Curriculum scaffold. Primary connections: Modules 1, 3, and 4.

11 IX - Computational physics methods

67. Basics 67: Discretization and Numerical Error
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

68. Basics 68: Finite-Difference Methods
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

69. Basics 69: Finite-Volume Methods
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

70. Basics 70: Finite-Element Methods
Curriculum scaffold. Primary connections: All numerical modules, especially Modules 3, 4, 6, 8, and 9.

12 X - Multiphysics coupling

13 XI - Physics-informed machine learning

14 XII - Graph neural networks and scientific graphs